

Yue Zhang

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EDUCATION

University of Waterloo

Bachelor of Mathematics in Data Science; GPA: 85/100(3.9/4.0)

Waterloo, ON, Canada

2020 - 2024

New York University Courant

Master of Science in Computer Science

New York, NY, United States

2024 - 2026(*expected*)

SKILLS SUMMARY

- **Interests:** Machine Learning; Cryptography(MPC, Differential Privacy, Federated learning)
- **Languages:** C++, C, Python, SQL, R, Racket
- **Likes:** Abstract Algebra, Graph Theory; Game Theory
- **Soft Skills:** Always positive mindset, Always open to new things

EXPERIENCE

New York University Langone Health

Research Assistant

New York, NY, United States

October 2024 - Present

Massachusetts Institute of Technology

Summer Exchange Program

Remote

July 2020 - Sep 2020

- **Courses:** Machine Learning plus Fintech
- **Project:** Build an Mortgage Lending Model with Convolutional Neural Networks (ConvNets)

PROJECTS

- **(Differential Privacy) Federated Learning:** Differential private federated learning (DPFL) algorithms and realizing protocol under single untrusted server
 - **Methods:** DP-SGD, DP-FTRL
- **(Transfer Learning) Adaptive Transfer Learning:** A knowledge transfer procedure focuses only on features, enables cross-architecture transfer from multiple pre-trained models
 - **Methods:** Bayesian Machine Learning, Gaussian Process
 - **Website:** https://hanbunko223.github.io/project/Adaptive_Transfer_Learning.pdf
- **(Machine Learning) Predicting Hit or Flop, A Classification Analysis of Spotify Music Dataset:** Classification models design and analysis to predict if a song is a hit or a flop
 - **Logistic Regression:** statistical inference
 - **Generative Classifiers:** LDA, QDA
 - **Support Vector Machine(SVM):** Polynomial Kernal, Radial Basis Function
 - **Tree-Based Methods:** Decision Tree, Random Forest
 - **Neural Networks:** Optimization of dimension and number of hidden layers
- **(Time Series Analysis) Prediction of Recent Toronto Average Temperature:** Statistical models design and analysis for time series data fitting
 - **Analysis:** Trend, Seasonality, Autocorrelation
 - **Smoothing:** Regression with regularization, the Holt-Winters algorithm, Differencing
 - **Random part:** Box-Jenkins
- **(C++ Programming)Biquadris Game Project:** Produce the video game Biquadris, a Latinization of the game Tetris
 - **Object-Oriented Programming:** design, test, debug, document a medium-sized program

HONORS AND AWARDS

- Third Prize in China National High School Mathematics League
- University of Waterloo President's Scholarship

VOLUNTEER EXPERIENCE

- **Volunteer Notetaker** Waterloo, ON
Take notes for classmate registered with AccessAbility Services Spring 2022

OTHERS

- **Nationwide Top20 in the game Hearthstone** Beijing, China
24Win-2Lose October 2019